AI-POWERED HELPDESK CHATBOT FOR FC INTERNAL QUALITY ASSURANCE (IQA)

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AI-POWERED HELPDESK CHATBOT FOR FC INTERNAL QUALITY ASSURANCE (IQA)

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A thesis submitted in fulfilment of the
requirements for the award of the degree of
Bachelor in Computer Science

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JULY 2025

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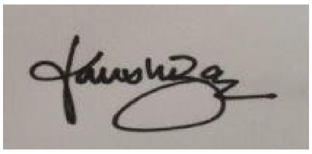
I declare that this thesis entitled *"AI-Powered Helpdesk Chatbot for FC Internal Quality Assurance (IQA)"* is the result of my own research except as cited in the references. The thesis has not been accepted for any degree and is not concurrently submitted in candidature of any other degree.

Signature : 

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ACKNOWLEDGEMENT

First and foremost, I would like to express my heartfelt gratitude to my supervisor, **Prof. Madya Ts. Dr. Rohayanti Binti Hassan**, for her unwavering support, guidance, and constructive feedback throughout the duration of this project. Her valuable suggestions, timely reviews, and continuous encouragement played a pivotal role in shaping the direction and outcome of this work.

This project was carried out independently as part of my industrial training, and her mentorship was instrumental in helping me overcome challenges and maintain focus across all stages of development—from initial design to final implementation.

I would also like to extend my appreciation to the faculty members who provided feedback during demonstration sessions, contributing practical insights that enhanced the quality and relevance of the system.

Lastly, I am deeply thankful to my family and close friends for their emotional support and motivation throughout this journey. Their belief in my abilities gave me the strength to persevere and complete this project successfully.

ABSTRACT

This industrial training project presents the design and development of an AI-powered document-aware QA chatbot tailored for academic and industrial environments. The system enables users to query the content of uploaded PDF documents—such as manuals or internal reports—through a natural language interface.

The project was developed as a full-stack web application using React.js for the frontend, Node.js with Express.js for the backend, MongoDB Atlas as the database, and Tailwind CSS for UI styling. Semantic chunking of PDF documents was achieved using a custom tokenizer, with vector embeddings generated via the all-MiniLM model served locally through Ollama. These embeddings were stored and queried using cosine similarity to retrieve the most relevant document segments.

Language inference was handled entirely through locally hosted LLMs integrated via the Ollama framework. The system supports role-based access (Visitor, Admin, System Admin), multi-turn conversations, persistent chat histories, and efficient context filtering to optimize relevance and response speed. Deployment was carried out in a controlled development environment, and the system was rigorously tested through API-level and user interface-level validations.

The outcome of this project demonstrates the viability of integrating scalable AI tools for intelligent document analysis and query handling. It bridges full-stack development with cutting-edge NLP techniques, providing a practical solution for real-time, document-aware AI assistance.

ABSTRAK

Projek latihan industri ini membentangkan reka bentuk dan pembangunan chatbot soal jawab (QA) berkuasa AI yang memahami dokumen, yang disesuaikan untuk persekitaran akademik dan industri. Sistem ini membolehkan pengguna bertanya soalan berdasarkan kandungan dokumen PDF yang dimuat naik — seperti manual atau laporan dalaman — melalui antara muka bahasa semula jadi.

Projek ini dibangunkan sebagai aplikasi web penuh (full-stack) menggunakan React.js untuk antaramuka pengguna, Node.js dengan Express.js untuk bahagian belakang (backend), MongoDB Atlas sebagai pangkalan data, dan Tailwind CSS untuk penataan UI. Penyenaraian semantik dokumen PDF dicapai melalui penyesuai tokenisasi tersuai, dengan penjanaan embedding vektor menggunakan model all-MiniLM yang dijalankan secara setempat melalui Ollama. Embedding ini disimpan dan dicari semula menggunakan persamaan kosinus untuk mendapatkan segmen dokumen yang paling relevan.

Inferens bahasa dikendalikan sepenuhnya melalui model bahasa besar (LLM) yang dihoskan secara setempat menggunakan Ollama. Sistem ini menyokong capaian berasaskan peranan (Pelawat, Admin, Sistem Admin), perbualan berbilang giliran, sejarah chat yang berterusan, dan penapisan konteks yang cekap untuk mengoptimumkan kerelevanan dan kelajuan respons. Pelaksanaan dibuat dalam persekitaran pembangunan terkawal dan sistem telah diuji dengan teliti melalui pengesahan tahap API dan antaramuka pengguna.

Hasil projek ini menunjukkan kebolehlaksanaan integrasi alat AI berskala untuk analisis dokumen pintar dan pengendalian pertanyaan. Ia menggabungkan pembangunan aplikasi penuh dengan teknik NLP terkini, menyediakan penyelesaian praktikal untuk bantuan AI masa nyata yang memahami dokumen.

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LIST OF ABBREVIATIONS

-

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CHAPTER 1

INTRODUCTION

1.1 Training Organization Overview

The industrial training was conducted under the Faculty of Computing, Universiti Teknologi Malaysia (UTM), located in Johor Bahru, Malaysia. As one of the core faculties in UTM, the Faculty of Computing focuses on nurturing highly skilled graduates in areas such as software engineering, artificial intelligence, cybersecurity, data science, and computer networks.

Problem Background

Core Business and Activities:

- Delivering undergraduate and postgraduate academic programs in Computer Science.
- Conducting industry-collaborative research in AI, machine learning, and computing systems.
- Developing enterprise-level applications and systems used by internal university departments.
- Providing students with access to real-world projects, simulations, and system deployments.

Staff Strength:

The faculty comprises more than 100 academic and technical staff members, including lecturers, professors, research fellows, and system developers. Each division is supported by lab assistants, postgraduate researchers, and undergraduate interns.

1.2 Organization Structure

The organizational structure of the Faculty of Computing is hierarchical and collaborative. It includes the following key units:

- **Dean's Office** – Administrative oversight and strategic direction
- **Academic Departments** – Software Engineering, Data Science, Multimedia and Graphics, Networks & Security
- **Research Groups & Labs** – AI & Machine Learning, Software Quality, Intelligent Computing
- **Technical Unit** – Manages system development, internal IT infrastructure, and digital services

My industrial training was supervised and coordinated by my supervisor at the Quality and Strategy Unit, Faculty of Computing, Universiti Teknologi Malaysia (UTM), located in N28A. I was provided with access to the necessary systems and hardware during the course of my training.

1.3 Project Objectives

I was attached to the Software Engineering and Intelligent System Lab. This unit is primarily responsible for:

- Designing and deploying an AI-powered document-aware chatbot system capable of semantic PDF content extraction, embedding, and intelligent retrieval using vector similarity techniques.
- Integrating locally hosted large language models (LLMs) via Ollama for real-time, multi-document natural language querying, and ensuring accurate and context-aware responses across multiple user roles.

- Developing a secure, full-stack web application with role-based authentication, interactive dashboards, and a responsive user interface using React.js and Tailwind CSS.

My core responsibility was to develop an AI-powered chatbot system that can intelligently answer industrial questions based on uploaded PDF documents. This project involved frontend and backend development, AI integration, user role-based access control, and testing workflows.

1.4 Planning Training Program

The following table outlines the planned training schedule, which was executed in close coordination with my supervisor:

Table 1.1 Planned Gantt Chart of Training Activities

Week	Activity Description
1	Orientation, environment setup, repo access
2	UI design (dark theme, layout, dashboard mockups)
3	Backend setup (Node.js, MongoDB, Express)
4	PDF upload, parsing and embedding integration
5	LLM integration via Ollama
6	Chat history + role-based access control
7	System testing, bug fixing, final polish
8	Documentation, code review, final report

This program was tailored to cover practical implementation, AI-based system integration, and project documentation under real working constraints.

1.5 Report Structure

This report is structured into four key chapters:

- **Chapter 1:** Introduction to the organization, structure, and training overview

- **Chapter 2:** Detailed explanation of the project tasks, technologies used, and outcomes
- **Chapter 3:** Reflection on personal development, skills gained, and team collaboration
- **Chapter 4:** Final conclusions, challenges, solutions, and suggestions for improvement